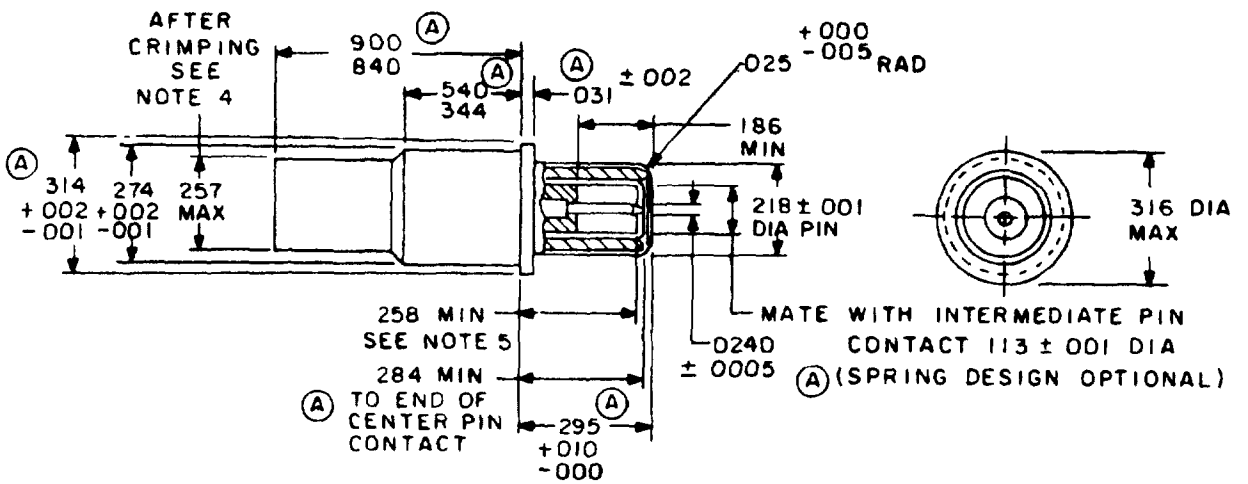
MILITARY SPECIFICATION SHEET

CONTACT, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, PIN, SIZE 8

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contact described herein shall consist of this specification and the latest issue of MIL-C-39029



INCHES	MM	INCHES	MM	INCHES	MM
.0005	0.013	.113	2.87	.344	8.74
.001	0.03	.166	4.22	.540	13.72
.002	0.05	.171	4.34	.587	14.91
.003	0.08	.218	5.54	.622	15.80
.005	0.13	.257	6.53	.625	15.88
.006	0.15	.274	6.96	.647	16.43
.023	0.58	.286	7.26	.840	21.34
.0240	0.610	.314	7.98	.900	22.86
.031	0.79	.316	8.03		

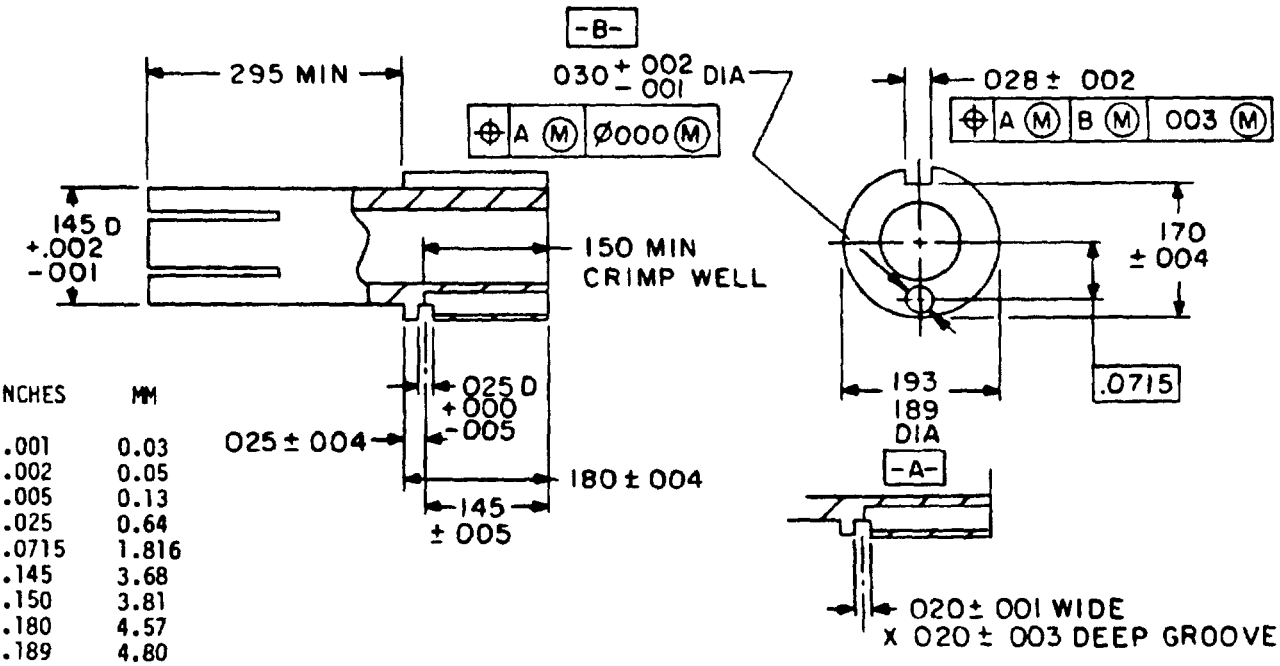
NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Diameter shall not exceed .276 (7.01 mm) over recovered heat shrink tubing.
5. Point at which a square ended pin of the same basic diameter as the mating contact first engages the intermediate contact spring. Provision for clearance hole shall be provided for the test pin.

Ⓐ FIGURE 1. Concentric twinax contact.

Ⓐ denotes changes

FSC 5935

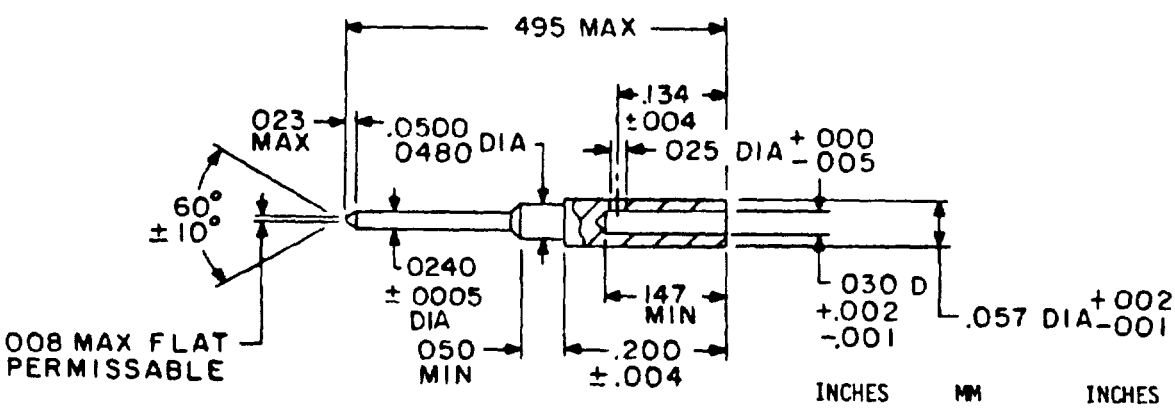
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INCHES	MM
.001	0.03
.002	0.05
.005	0.13
.025	0.64
.0715	1.816
.145	3.68
.150	3.81
.180	4.57
.189	4.80
.193	4.90

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.

Ⓐ FIGURE 2. Intermediate socket contact.



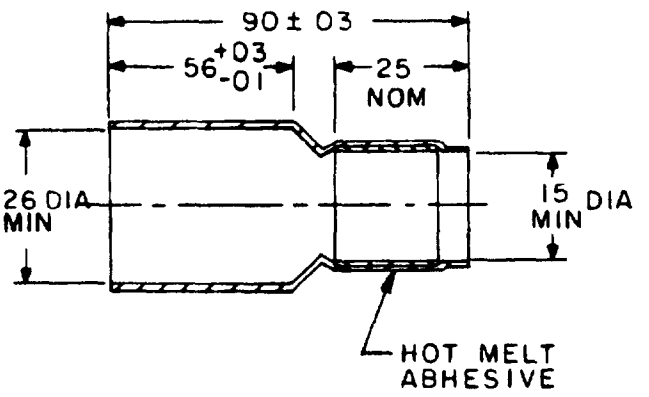
INCHES	MM	INCHES	MM
.0005	0.013	.025	0.64
.000	0.00	.030	0.76
.001	0.03	.0480	1.219
.002	0.05	.0500	1.270
.004	0.10	.050	1.27
.005	0.13	.057	1.45
.006	0.15	.134	3.40
.008	0.20	.147	3.73
.023	0.58	.200	5.08
.024	0.61	.495	12.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.

Ⓐ FIGURE 3. Center pin contact.

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INCHES	MM
.00	0.
.010	0.25
.03	0.8
.15	3.8
.25	6.4
.56	14.2
.90	22.9



NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

Ⓐ FIGURE 4. Heat shrink boot.

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REQUIREMENTS

(A) Qualification Contacts shall comply with reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999

Design and construction

Dimensions and configuration See figures 1 through 4 and table I

TABLE I. Design characteristics

BIN code	Color bands			Contact cavity size	Cable accommodated	Type	Class
	1st	2nd	3rd				
(A) 529	Green	Red	White	8	M17/176-00002	D	B

(A) Assembly procedure Manufacturer's recommended assembly instructions shall be shipped with unit package.

(A) Material

Heat shrink boot Shall be in accordance with MIL-I-23053/8

Electrical

Low signal level contact resistance (center and intermediate contacts only) See Table II

Contact resistance See table III.

Frequency 0 to 20 MHz (operating frequency range)

Voltage rating 500 volts rms maximum, working voltage at sea level, 125 volts rms maximum at 70,000 feet

TABLE II Low signal level contact resistance (center and intermediate contacts only)

BIN code	Cable accommodated	Maximum contact resistance (milliohms)	
		Initial	After conditioning
(A) 529	M17/176-00002	55	66

TABLE III. Contact resistance.

BIN code	Contact	Cable accommodated	Test current (amperes)	Maximum voltage drop (millivolts)		
				25°C, +3°C, -0°C		+175°C, +3°C, -0°C
				Initial	After conditioning	After conditioning
(A) 529	Center	M17/176-00002	1.0	55	66	94
529	Intermediate	M17/176-00002	1.0	55	66	94
529	Outer	M17/176-00002	12.0	75	90	128

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(A) Dielectric withstanding voltage Shall be as specified in table IV

TABLE IV Dielectric withstanding voltage.

Contacts	Altitude	Test voltages ac rms
Center to intermediate	Sea level	1000
Intermediate to outer	Sea level	500

(A) Mating contact. Shall be in accordance with MIL-C-39039/91.

Mechanical

(A) Contact engagement and separation force (socket contacts only) The engagement depth shall be a minimum of 0.7 of the minimum socket bored. The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table V. Provision for clearance hole on outer contact. Test pins shall be provided. The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table V.

(A) TABLE V. Contact engagement and separation force.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.1140 +.0000 -.0001	NA	NA	18	22	NA
.1120 +.0001 -.0000	0.5	0.4	NA	NA	NA

Crimp tensile strength (center, intermediate, and outer contact crimp joint). See table VI.

(A) TABLE VI. Crimp tensile strength (at ambient).

BIN code	Cable accommodated	Axial load (pounds, minimum)		
		Center contact	Intermediate contact	Outer contact
529	M17/176-00002	8	8	25

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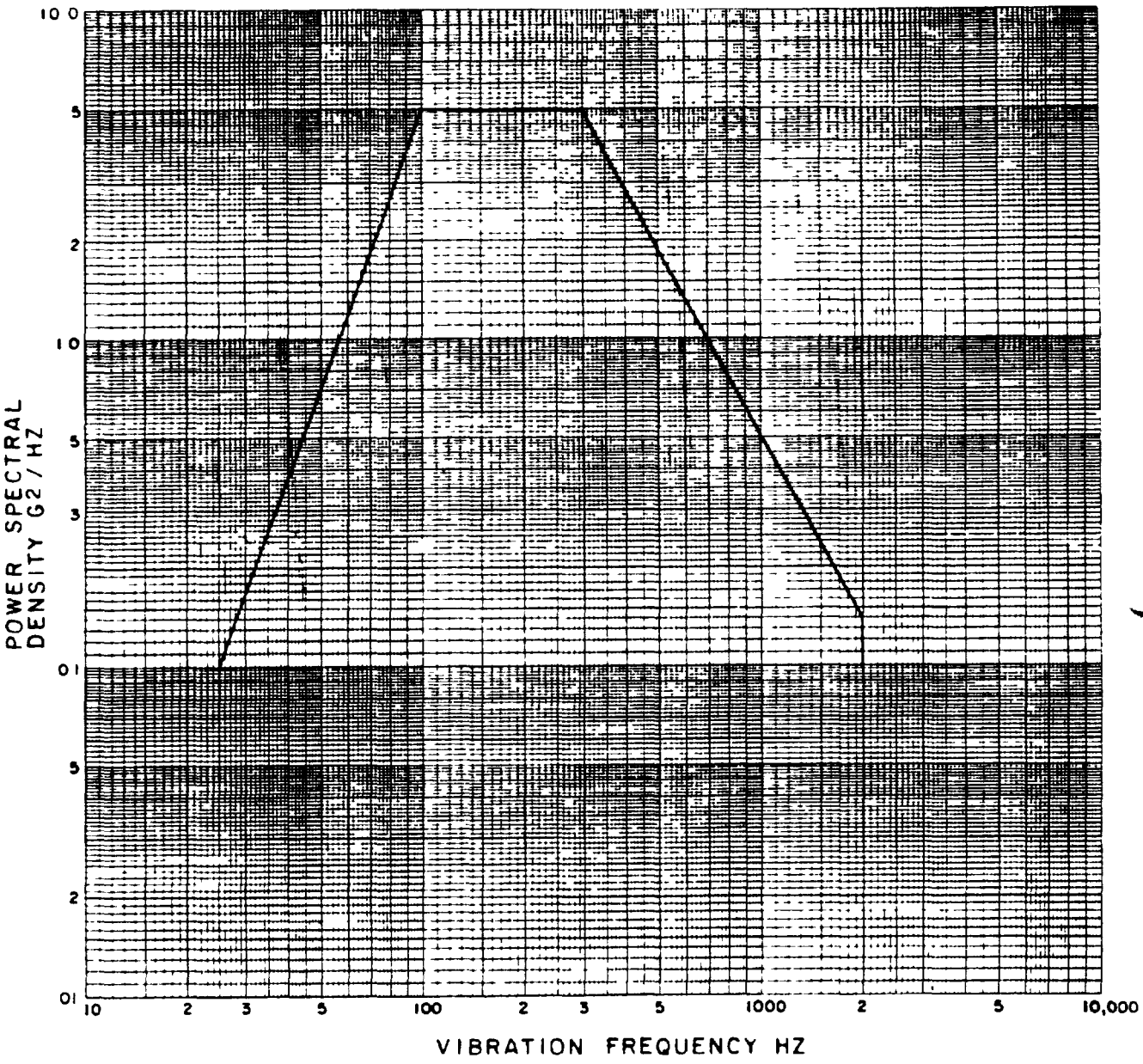


FIGURE 5. Vibration envelope.

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Vibration Method 2005 of MIL-STD-1344, test condition V The following details shall apply

(A) a Use the vibration envelope shown on figure 5.

b Vibration To be conducted at standard test conditions.

c Duration 8 hours in the longitudinal direction and 8 hours in a perpendicular direction (16 hours)

High impact shock All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during shock Connectors shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used Wired and mated connectors shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions Mounting fixture shall be in accordance with MIL-S-901, lightweight. The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914.4 mm).

a. Coupling Coupled by normal mounting means.

b. Test condition A Plug shall be terminated with at least 80 percent of wired contacts The wire bundle shall be provided with straight, open frame, strain relief accessory hardware

Tools Shall be as specified in tables VII and VIII

(A) TABLE VII. Tools (installing and removal).

BIN code	Installing and removal tool
529	M81969/14-06

TABLE VIII Tools (crimping).

	BIN code	Center contact tooling		Intermediate contact tooling		Outer crimp sleeve tooling	
		Basic crimping tool	Contact positioner 1/	Basic crimping tool	Die part no 1/	Basic crimping tool	Die part no 1/
(A)	529	M22520/2-01	K709	M22520/5-01	Y631 Die Closure B	M22520/5-01	Y631 Die Closure A

Environmental

(A) Temperature rating -65°C to +175°C

(A) Humidity Method 1002, type II, per MIL-STD-1344

1/ Daniels Manufacturing Corporation part number is specified Tools may be provided by other vendors if they are determined to be equivalent by the Qualifying Activity

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Part number See table IX

TABLE IX Part number.

BIN code	Part number	Supersedes
529	M39029/90-529	---

QPL evaluating activity Naval Avionics Center, D/714, 6000 East 21st Street, Indianapolis, Indiana 46218

Custodians
Army - CR
Navy - AS
Air Force - 85

Review activities
Navy - EC, SH, MC
Air Force - 11, 15, 17, 99
DLA - ES

User activities
Army - AT
Navy - OS

Agent:
DLA - ES

Preparing activity.
Navy - AS

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